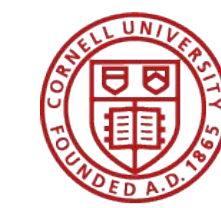


SpikeGPT: AN EFFICIENT Neuromorphic LLM



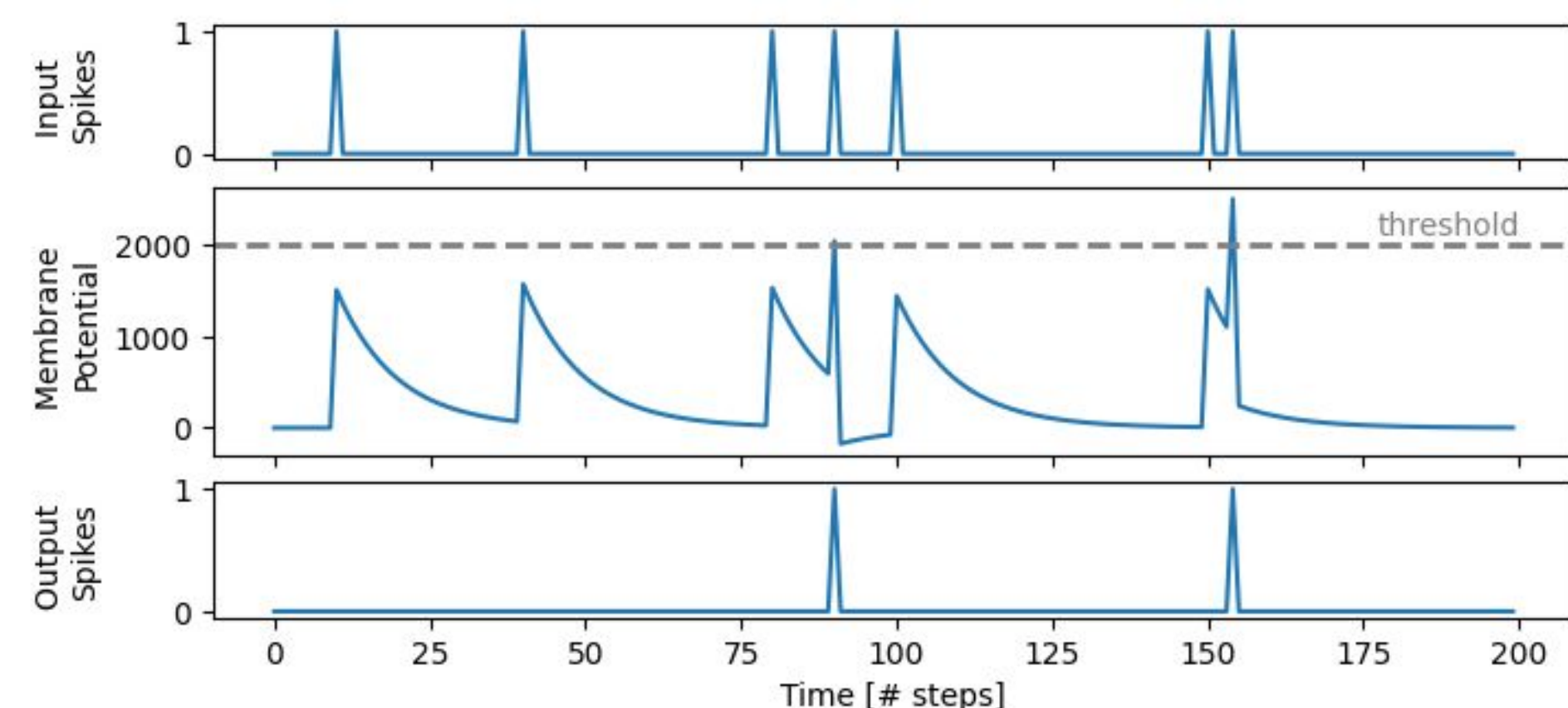
CornellBowers
Computer Science

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Spiking Neural Networks (SNN)

- SNNs communicate using discrete “spikes” - information encoded in **timing** and **frequency**
- Each neuron has an internal “voltage” and mimics three features of biological neurons:
 - State **exponentially decays** back to 0
 - Incoming spikes **integrate** state by set weights
 - When voltage passes a threshold, the neuron **fires** an output spike and resets its state
- MatMul replaced by addition, reducing computational cost



$$U[t] = \beta U[t - 1] + WX[t] - \theta S_{out}[t]$$

Methodology

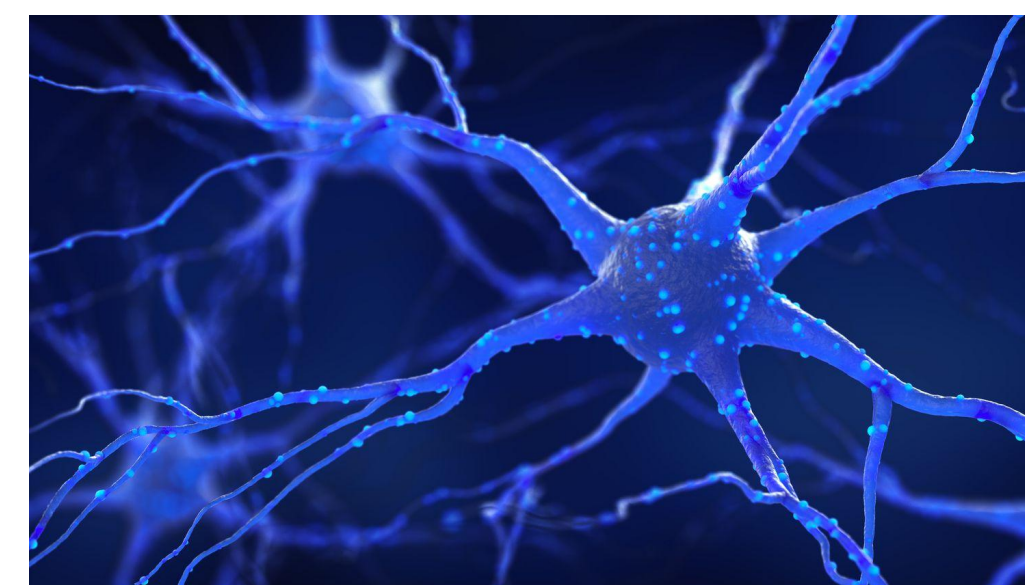
- Training Setup
 - Pretrained on language modeling datasets (Enwik8)
 - Next-character prediction objective (NLG)
 - Fine-tuned for sentiment classification tasks (NLU)
- Energy Evaluation - use cost estimates per operation
 - Standard GPT: dominated by multiply-accumulates (4.5 pJ)
Energy $\approx \#MAC \times MAC \text{ cost}$
 - SpikeGPT: dominated by cheaper accumulates (0.9 pJ),
Energy $\approx \#accum \times accum \text{ cost} \times \text{spike firing rate}$

2	-1	4	0	3	-2	1	5	-3	2
-3	2	1	7	-2	0	4	-1	3	1
0	5	-2	1	2	-3	1	0	-2	4
1	-4	3	2	0	-1	2	2	1	-2
2	0	-1	4	1	-2	3	1	0	2
-1	3	2	-2	4	1	0	-3	2	1
3	-2	0	1	-1	2	4	0	1	-3
1	2	-3	0	2	-1	1	3	-2	0

1	2	-1	0	3	-2	1	4
-2	0	3	-1	2	0	0	-3
4	-1	0	2	1	0	-2	1
0	3	-2	1	2	-1	3	0
1	-2	1	0	0	2	-1	2
-3	1	2	-1	3	0	1	-2
2	0	-1	3	-2	1	0	3
1	2	0	-2	1	3	-1	0
-1	1	3	0	-1	2	-2	1
3	-2	1	1	0	-3	2	-1

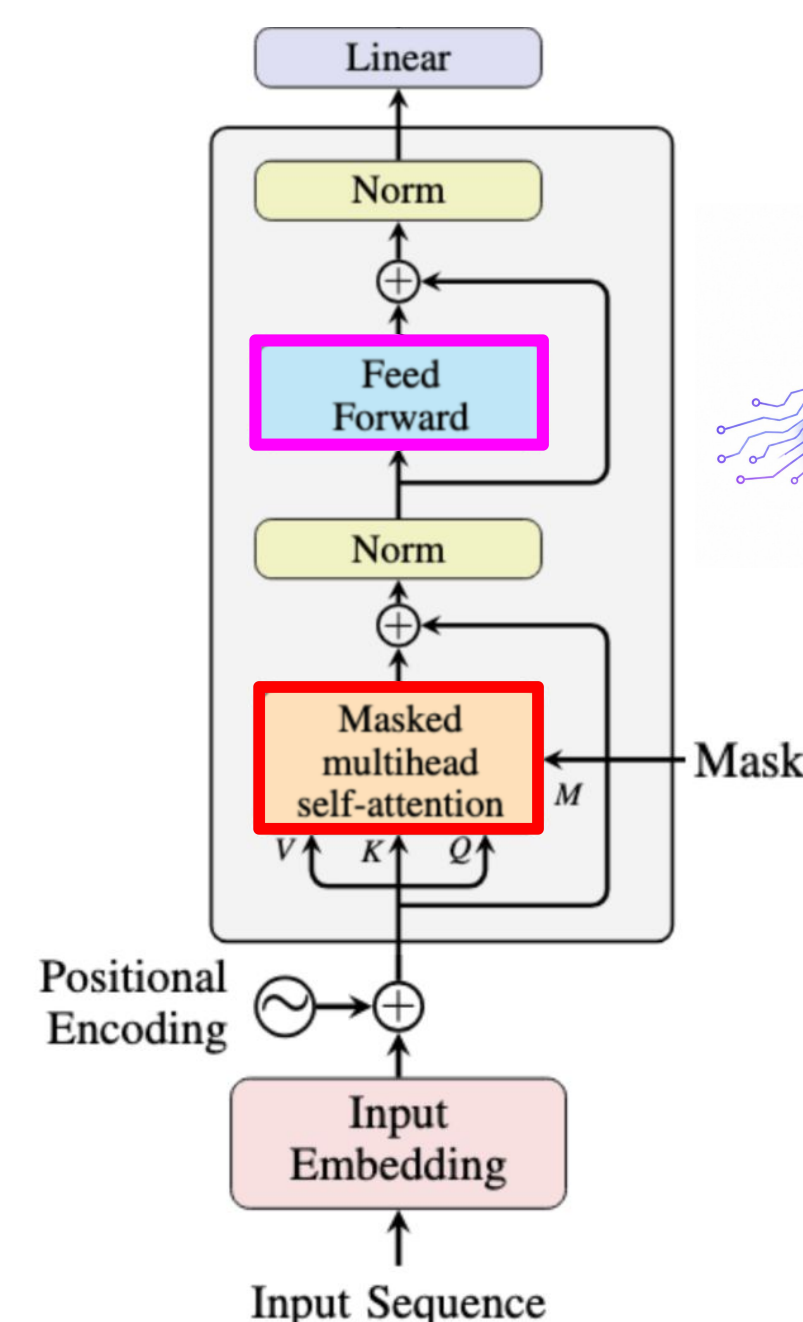
Lots of Multiplies,
Lots of Power!

No Multiplies,
Much Less
Power!

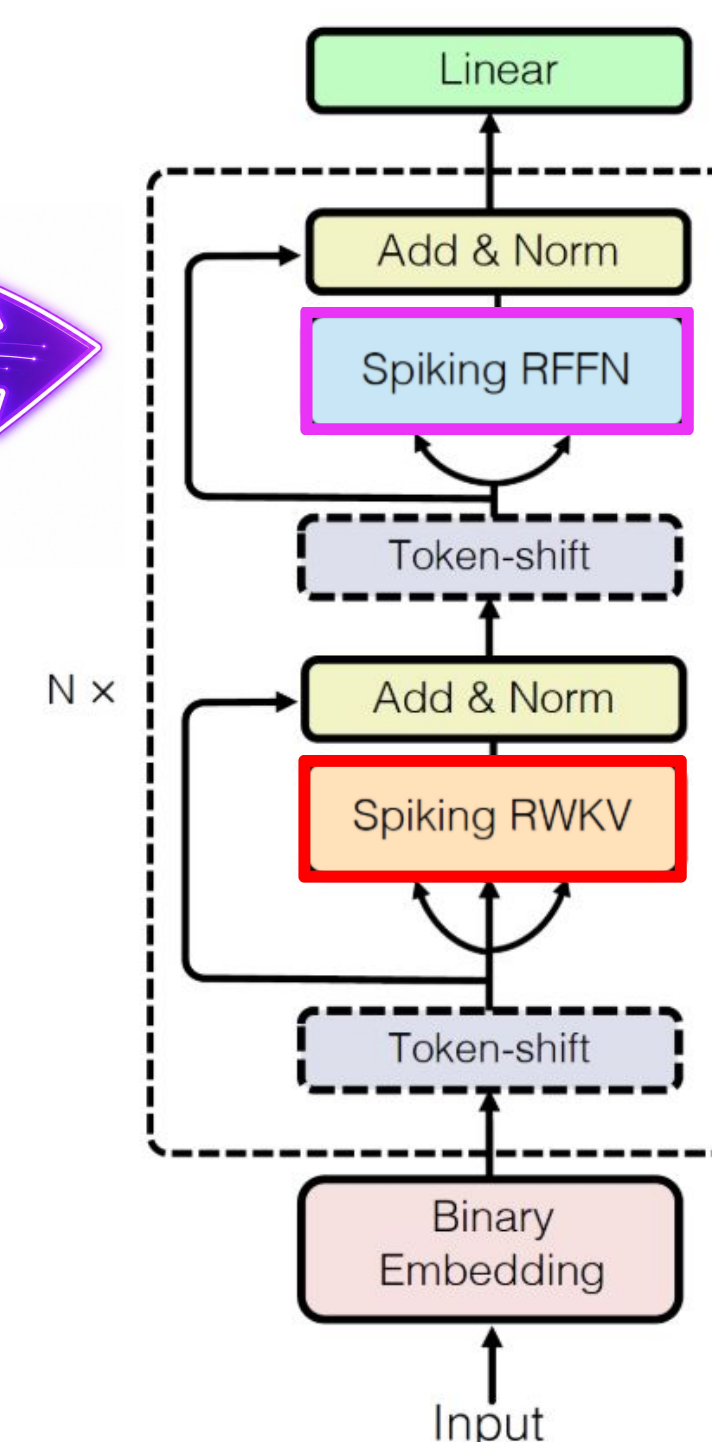


Architecture

Standard GPT

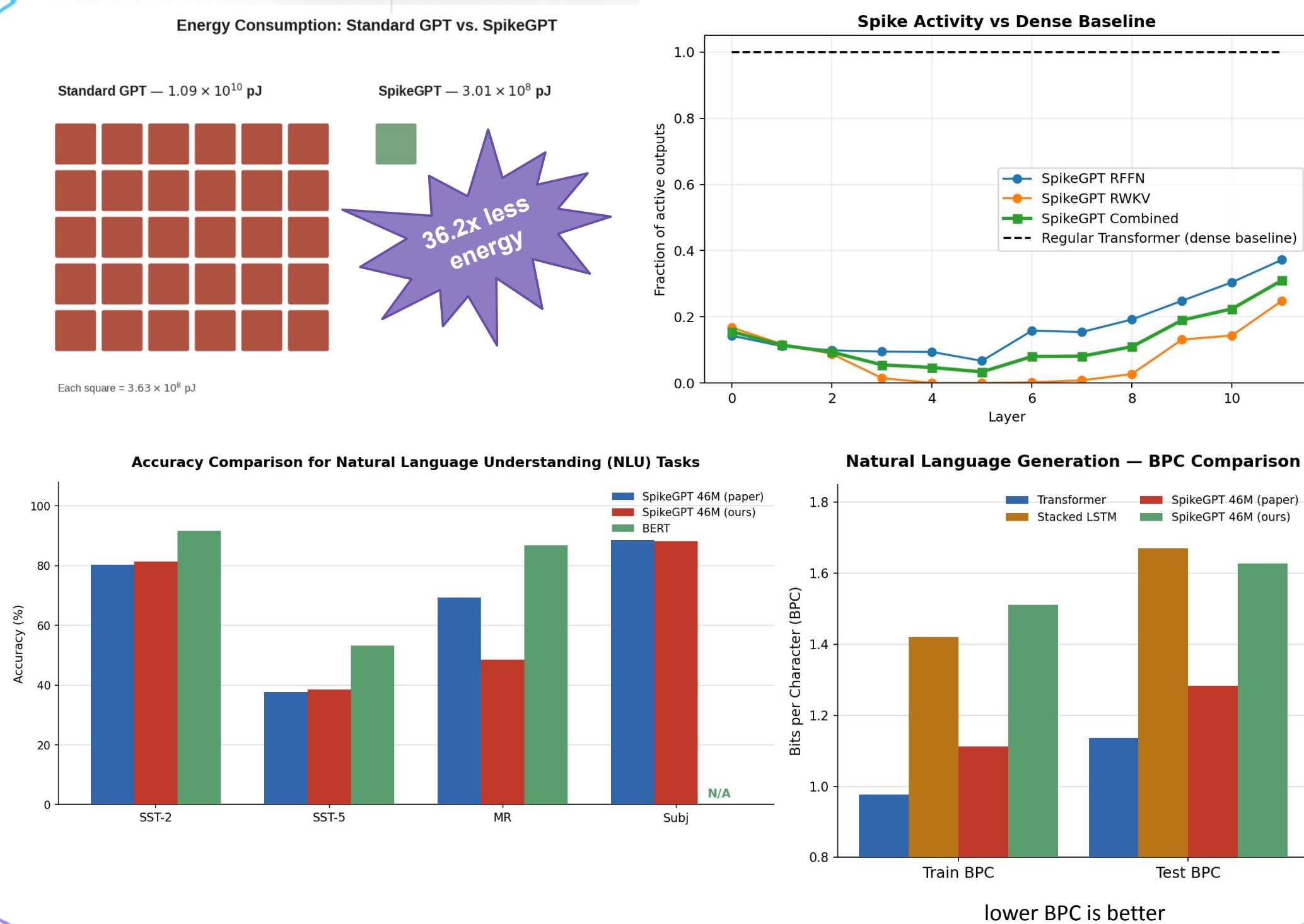


SpikeGPT

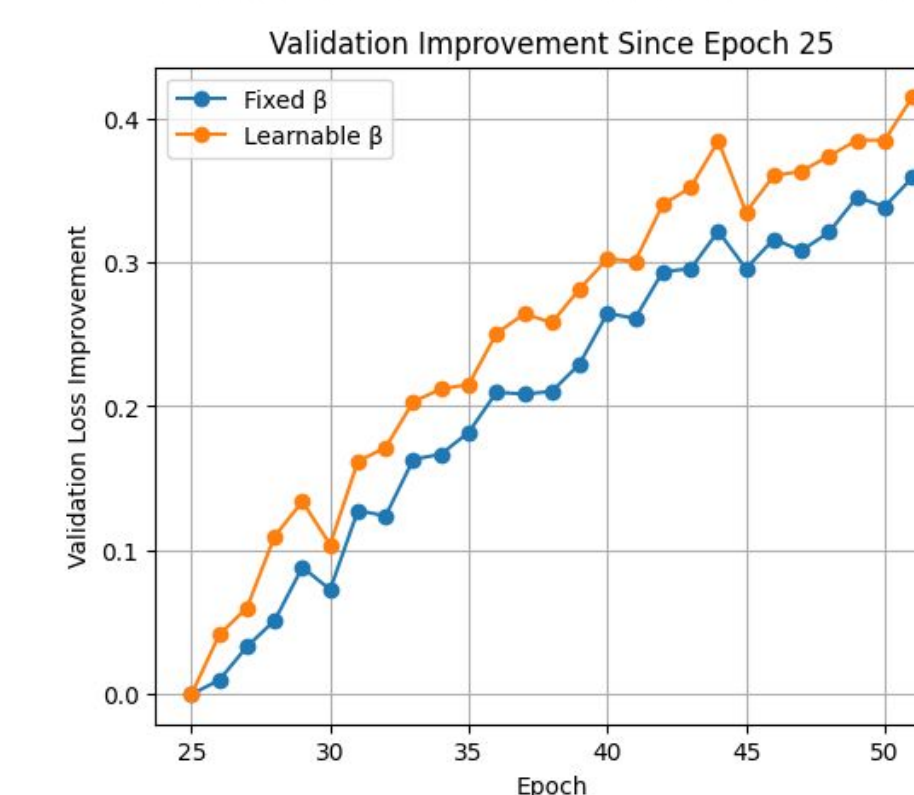


- Replace feed forward with SRFFN (MLP with spiking neurons)
- Replace $O(n^2)$ self-attention with $O(n)$ spiking RWKV (recurrent, exponentially weighted sum of past tokens)

Results



Exploration



- Learnable β parameter, system learns how long to “remember” spikes for
- Trained in phases: first with fixed β , then with learnable β
- Swap at early epoch (25) showed best improvement
- Train fully with learnable β ?

References

- [1] Zhu, R. et al. (2022). SpikeGPT: Generative Pre-trained Language Model with Spiking Neural Networks. Transactions on Machine Learning Research (TMLR). <https://arxiv.org/abs/2302.13939>
- [2] Cornell University Department of Computer Science. Weinberger, K. et al. CS 4782 Deep Learning. www.cs.cornell.edu/courses/cs4782/2026sp/

